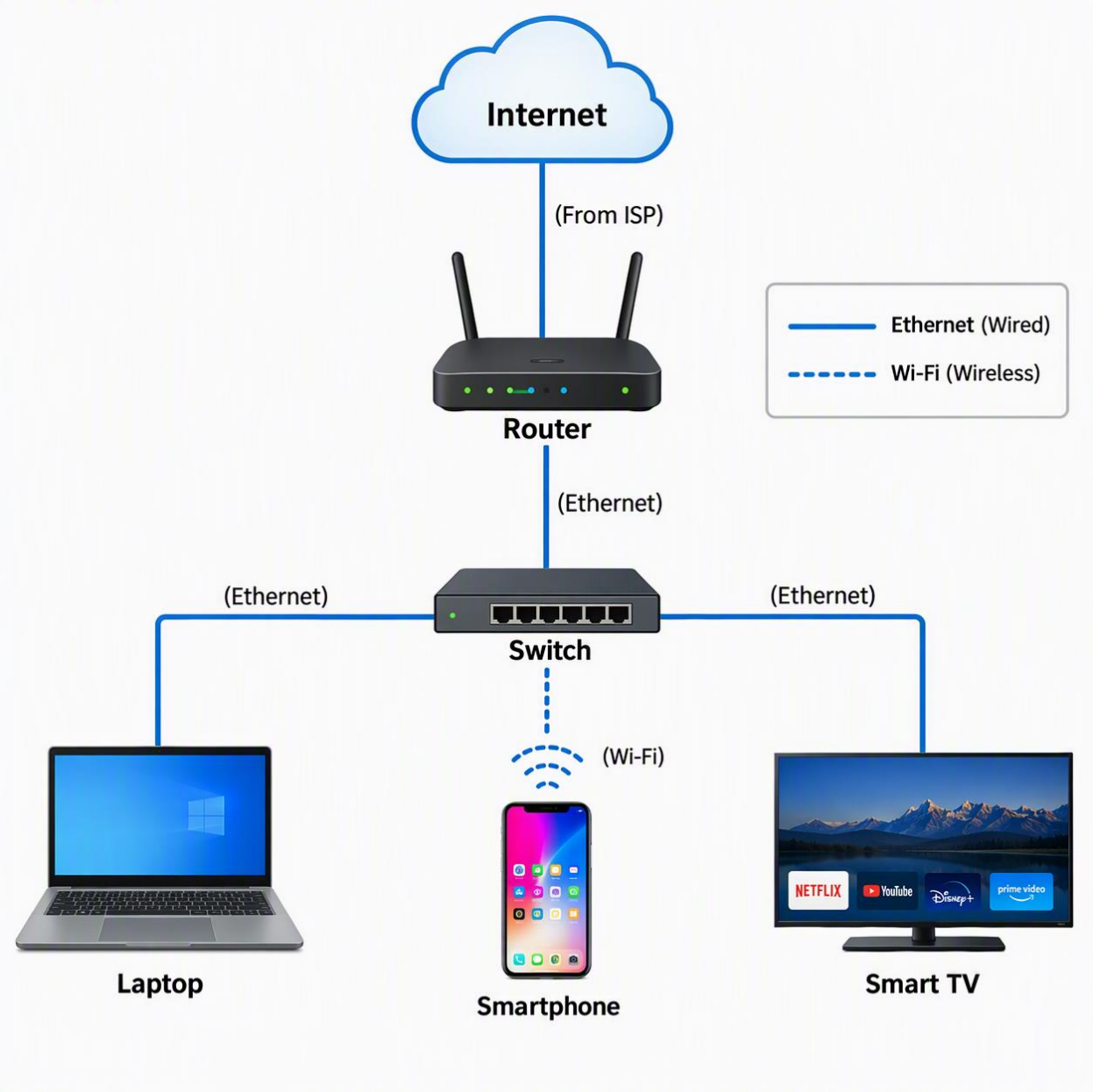
1 Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



2 Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Network Device	Main Function	Real-World Use Case
Router	Connects different networks and provides internet access.	I use a Wi-Fi router to access the internet and watch videos on YouTube.
Switch	Connects multiple devices within a LAN.	In a computer lab, a switch connects computers for file sharing and LAN gaming.
Hub	Connects devices and broadcasts data to all connected ports.	In an old computer lab, a hub can connect computers for basic local network communication.
Access Point (AP)	Provides wireless network access to devices.	I use an access point to connect my smartphone and laptop to Wi-Fi for WhatsApp, Instagram, and online classes.

3 Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Scenario: Ordering Food on Zomato

When I order food on Zomato using my smartphone, my phone connects to Wi-Fi through an Access Point (AP). The data then travels through the Router, which forwards my request to the internet. A Switch helps transfer data between devices within the local network and connects the network to other equipment. The request reaches Zomato's server, which processes my order and sends confirmation back to my phone through the internet.

Thus, the router, switch, and access point help me place my food order online.

4 Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

Two Devices to Connect a Gaming PC to the Internet

1. Powerline Adapter

- It uses the existing electrical wiring in your home to carry network data.
- Connect one adapter to the router using an Ethernet cable and plug it into a power socket.
- Plug the second adapter near the gaming PC and connect it to the PC using an Ethernet cable.
- Use: Provides a wired connection for online gaming.

2. Wi-Fi Extender with an Ethernet Port

- It receives the Wi-Fi signal from the router and extends its coverage.
- Place it near the gaming PC and connect the PC to the extender using an Ethernet cable.
- Use: Allows the PC to access the internet without running a long Ethernet cable from the router.

Conclusion: Both devices can connect the gaming PC to the internet. A powerline adapter may provide a more stable gaming connection, depending on your home's electrical wiring.